```
print("Creating a Simple Calculator")

a=float(input("Please Enter First no:"))

b=float(input("Please Enter Second no:"))

print("Choose opertion:+,-,*,/")
print("For Exit press Enything....")
c=str(input())

if(c=="+"):
    print(a+b)
elif(c=="-"):
     print(a-b)
elif(c=="*"):
     print(a*b)
elif(c=="/"):
     print (a/b)
else: print("You pressed exit")
```

Command Prompt

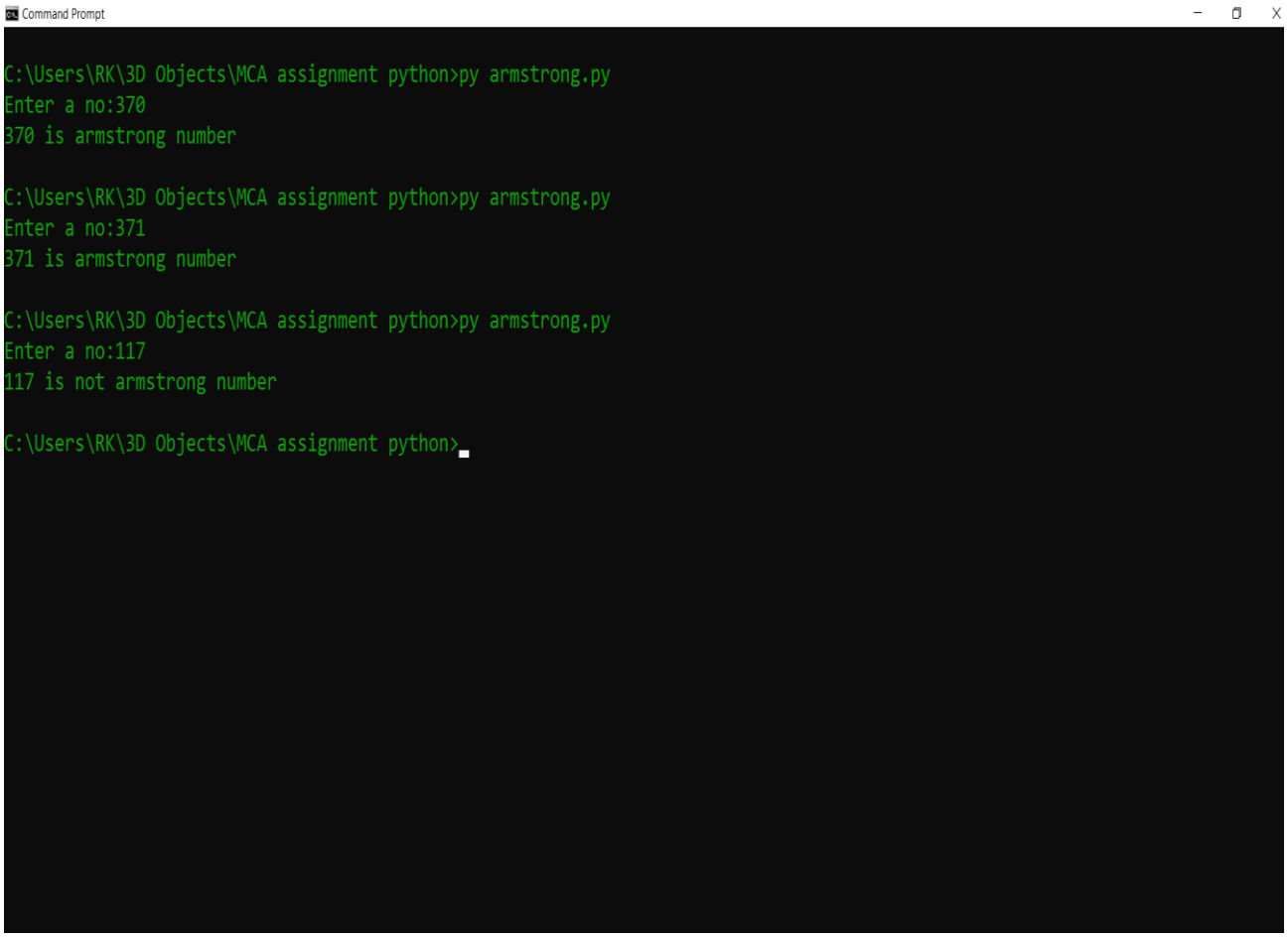
```
C:\Users\RK\3D Objects>py calcu.py
Creating a Simple Calculator
Please Enter First no:256
Please Enter Second no:25
Choose opertion:+,-,*,/
For Exit press Enything....
+
281.0

C:\Users\RK\3D Objects>py calcu.py
Creating a Simple Calculator
Please Enter First no:89.23
Please Enter Second no:17.23
Choose opertion:+,-,*,/
For Exit press Enything....
-
72.0

C:\Users\RK\3D Objects>py calcu.py
Creating a Simple Calculator
Please Enter First no:57
Please Enter Second no:558
Choose opertion:+,-,*,/
For Exit press Enything....
*
31806.0

C:\Users\RK\3D Objects>py calcu.py
Creating a Simple Calculator
Please Enter First no:56
Please Enter Second no:8
Choose opertion:+,-,*,/
```

```
a=int(input("Enter a no: "))
sum=0
t=a             #t=temp. a number
while (t>0):
    digit=t%10
    sum =sum+ (digit**3)
    t//=10
if a==sum:
    print ("{0} is armstrong number".format (a))
else:
    print ("{0} is not armstrong number".format (a))
```



Experiment No :4

Experiment name:Write a python program to print the prime no(using the function)

Program:

```
def prime(n):
    if(n>1):
            for i in range(2,int(n/2)+1):
                if(n%i)==0:
                 print("{0}: is not an prime no".format(n))
                 break
            else:
                 print("{0}: is a prime no".format(n))
    else:
        print("{0}: is not a prime no".format(n))


no =int(input("enter an number:"))
prime(no)
```

output:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\Rushikesh Kore MCA -1> python prime.py
enter an number:3
3: is a prime no
PS D:\Rushikesh Kore MCA -1> python prime.py
enter an number: 4
4: is not an prime no
PS D:\Rushikesh Kore MCA -1>
```

Experiment No :5

Experiment name:Write a python program to count Vowels and Consonants

---

Program:

```
#Program to count Vowels and Consonants of a String

vcount = 0

ccount = 0


str = str(input("Enter a string"))

#Converting entire string to lower case to reduce the
comparisons str = str.lower() for i in range(0,len(str)):

#Checks whether a character is a vowel if str[i] in
('a',"e","i","0","u"):

str=str.lower()

for i in range(0,len(str)):

    if str[i] in ( "a","e","i","o","u"):

        vcount = vcount + 1

    elif (str[i]>='a' and str[i] <= 'z'):

        ccount = ccount + 1


print("number of vowel:",vcount)

print("number of  consonant are:",ccount)
```

output:

```
PS D:\Rushikesh Kore MCA -1> python voels.py
Enter a string This is a Python
number of vowel: 4
number of  consonant are: 9
PS D:\Rushikesh Kore MCA -1>
```

Program:

```
a=[]
ele=int(input("please enter Total Number of elments: "))
for i in range(1,ele+1):
     val=int(input("Please enter a value of %d Element: "%i))
     a.append(val)
small=large=a[0]
minele=maxele=0
for j in range(1,ele):
     if(small > a[j]):
          small=a[j]
          minele=j
     if(large < a[j]):
          large=a[j]
          maxele=j
print("-------------------------------------------------")
print("The Smallest Elements in List is: ",small)
print("The Largest Elements in List is: ",large)
print("index possition of largest element in list: ",maxele)
print("index possition of smallest element in list: ",minele)
```

Output:

```
please enter Total Number of elments: 5
Please enter a value of 1 Element: 10
Please enter a value of 2 Element: 25
Please enter a value of 3 Element: 70
Please enter a value of 4 Element: 48
Please enter a value of 5 Element: 50
-------------------------------------------------
The Smallest Elements in List is:  10
The Largest Elements in List is:  70
index possition of largest element in list:  2
index possition of smallest element in list:  0
```

Program:

```
n=int(input("Enter a total no of values:"))
d={}
for i in range(n):
    key=int(input("Enter Keys:"))
    Values=int(input("Enter Values:"))
    d[key]=Values
print(d)

sum=0
for i in d:
    sum=sum+d[i]
print("Sum of Element in dictionary is:",sum)
```

Output:

```
Enter a total no of values:3
Enter Keys:0
Enter Values:10
Enter Keys:1
Enter Values:20
Enter Keys:2
Enter Values:30
{0: 10, 1: 20, 2: 30}
Sum of Element in dictionary is: 60
>
```

Program:

```
# creatting two sets
set1=set()
set2=set()
#adding element to set set1
for i in range(1,6):
    set1.add(i)

#adding element to set set2
for i in range(3,8):
    set2.add(i)

print("Set1=",set1)
print("Set2=",set2)
print("\n")

#Union of set1 and Set2
set3=set1|set2            #union =set1.union(set2)
print("Union of set1 and set2:set3=",set3)

#Intersection of set1 and Set2
set4=set1 & set2          # Intersection=set1.intersection(Set2)
print("Union of set1 and set2:set4=",set4)
print("\n")
```

```
#checking reletion between set3 and set4
if set3>set4:  #set3.issuperset(set4)
    print("set3 is superset of set4")
elif set3<set4:  #set3.issubset(set4)
    print("set3 is subset of set4")
else:
    print("set 3 is same as set4")

#displaying reletion between set4 and set3
if set4<set3:  #set4.issubset(set3)
    print("set4 is subset of set3")
    print("\n")
```

Output:

```
Set1= {1, 2, 3, 4, 5}
Set2= {3, 4, 5, 6, 7}


Union of set1 and set2:set3= {1, 2, 3, 4, 5, 6, 7}
Union of set1 and set2:set4= {3, 4, 5}


set3 is superset of set4
set4 is subset of set3


>
```

Program :

```
print("---------Pattern 1----------")
rows = 6
for i in range(rows):
    for j in range(i):
        print(i, end=' ')
    print('')

print("\n--------Pattern 2--------")
rows = 5
b = 0
for i in range(rows, 0, -1):
    b += 1
    for j in range(1, i + 1):
        print(b, end=' ')
    print('\r')

print("----------Pattern 3-------")
rows = 5
k = 2 * rows - 2
for i in range(rows, -1, -1):
    for j in range(k, 0, -1):
        print(end=" ")
    k = k + 1
    for j in range(0, i + 1):
        print("*", end=" ")
    print("")
```

Output:

Command Prompt

```
C:\Users\RK\3D Objects\python>python assignment_9.py
---------Pattern 1----------

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

--------Pattern 2--------
1 1 1 1 1
2 2 2 2
3 3 3
4 4
5
----------Pattern 3-------
        * * * * * *
         * * * * *
          * * * *
           * * *
            * *
             *

C:\Users\RK\3D Objects\python>
```

Program:

```
# star square
a=int(input("please enter a no: "))
print("print square star pattern")
for i in range(a):
    for i in range(a):
        print("*",end=" ")
    print()
```

Output:

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct  2 2023, 13:03:39) [MSC v.1935 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================= RESTART: D:\Rushikesh Kore MCA -1\pattern2.py ================
---------Pattern 4--------
Please enter a no :5
print square star pattern:
*****
*****
*****
*****
*****
>>>
```

Program:

```
import os

try:
    upperCount = 0
    lowerCount = 0

    # Get the directory of the script
    script_dir = os.path.dirname(os.path.abspath(__file__))

    # Use os.path.join to create a valid file path
    file_path = os.path.join(script_dir, "rk.txt")

    with open(file_path, "r") as f:
        while True:
            data = f.read(1)

            if not data:
                break

            if data.isupper():
                upperCount += 1
            elif data.islower():
                lowerCount += 1

            print(data, end="")

    print("\nTotal Upper case:", upperCount)
```

```
    print("Total lower case:", lowerCount)

except FileNotFoundError as e:
    print(e)

finally:
    try:
        f.close()
    except NameError:
        pass  # f was not defined
    print("ok done...!")
```

Output:

Command Prompt

```
C:\Users\RK\3D Objects\python>python assignment_10_python.py
HELLO PYTHON
my name is rk
Total Upper case: 11
Total lower case: 10
ok done...!

C:\Users\RK\3D Objects\python>
```